Quantifying Procedural Gradient Noise Quality and Implicit Geometry via Analytic Derivatives: A Theoretical and Empirical Monograph

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Abstract

Procedural gradient noise functions—spanning classical cubic-lattice Perlin noise, simplicial tessellations (Simplex), and patent-free variants (OpenSimplex2F, OpenSimplex2S)—serve as the foundational primitives of stochastic texture synthesis, terrain modeling, and volumetric rendering. Despite four decades of widespread adoption, the quantitative evaluation of noise quality has remained largely restricted to Fourier power spectra and qualitative visual inspection. In this work, we introduce a unified differential-geometric and statistical framework for evaluating procedural gradient noise and its geometric applications, enabled by exact, single-pass analytic derivatives implemented in the `ForgeNoise` architecture. Across seven reproducible numerical investigations (100% deterministic to four decimal places), we establish five primary theoretical and empirical contributions, alongside two foundational analyses for non-Euclidean and four-dimensional spaces:

1. **Exact Domain-Warp Folding Thresholds:** We prove that two-field domain warping $\mathbf{q}(\mathbf{p}) = \mathbf{p} + s\,\mathbf{w}(\mathbf{p})$ ceases to be a diffeomorphism at an exact, seed-dependent critical warp strength s_{crit} governed by the pointwise quadratic Jacobian determinant $\det \mathbf{J}(\mathbf{p}, s) = 1 + a(\mathbf{p})s + b(\mathbf{p})s^2$. Across 100 independent realizations of 2D Simplex noise, we establish $s_{\text{crit}} = 0.1252 \pm 0.0074$ ($q_{05} = 0.114$, $q_{50} = 0.129$, $q_{95} = 0.134$). Consequently, standard library defaults ($s = 1.0$) overshoot the injectivity limit by $8\times$, inducing severe topological folding. Furthermore, for N-octave fractional Brownian motion (fBm) warps with $N \geq 4$, we derive from quadrature gradient summation that $s_{\text{crit}}(N) \propto N^{-0.5}$ and measure an empirical scaling exponent of $\alpha = 0.544$ (8.9% relative error).
2. **Critical-Point Topology and Kac–Rice Discrimination:** Using Newton–Raphson refinement on the analytic gradient field $\nabla n(\mathbf{x}) = \mathbf{0}$, we demonstrate that all evaluated noise families obey the universal Kac–Rice and Morse-theoretic Gaussian prediction for the saddle-to-extremum ratio ($R_{\text{saddle}} \approx 1.975$ vs. 2.000, within 1.5%). Conversely, total critical-point density ρ_{cp} acts as a sharp discriminator of spectral bandwidth and lattice packing, separating Perlin (3.07 ± 0.13 unit^{-2}) from Simplex (6.61 ± 0.10 unit^{-2}) and OpenSimplex2F (6.83 ± 0.08 unit^{-2}) by a factor of $2.2\times$.
3. **Structure-Tensor Lattice Fingerprinting:** By integrating the local outer product of gradients $\mathbf{T} = \langle \nabla n \otimes \nabla n \rangle$ over circular windows, we simultaneously quantify local anisotropy via the eigenvalue ratio λ_2/λ_1 and extract global crystallographic symmetry via circular Rayleigh harmonics (R_4, R_6). The (R_4, R_6) phase space completely separates square-lattice algorithms (Perlin: $R_4 = 0.295, R_6 = 0.033$) from triangular/simplicial lattices (Simplex, OpenSimplex2F, OpenSimplex2S: $R_4 \in [0.063, 0.098], R_6 \in [0.170, 0.235]$) with 3–7$\times$ margin.
4. **Ridged Multifractal Anti-Percolation:** Testing the hypothesis that ridged multifractals ($1 - |n|$) undergo a connectivity phase transition above a critical persistence p_c, we report an informative negative result: across $p \in [0.30, 0.70]$, no percolation threshold exists. Instead, high-frequency octave injection monotonically

fragments the ridge skeleton, shrinking the largest connected component fraction by $5\times$ ($0.036 \to 0.007$) and tripling isolated component counts ($771 \to 2256$).

5. **Implicit Heightmap Raycasting via Free Analytic Gradients:** Formulating oblique ray-terrain intersection as root-finding on $F(t) = y(t) - H\,n(x(t), z(t))$, we demonstrate that Newton iteration driven by co-evaluated analytic derivatives requires $4.91\times$–$8.17\times$ ($6.54\times$ mean) fewer noise evaluations than bisection and $2.31\times$ fewer evaluations than finite-difference Newton across tolerances $\varepsilon \in \{10^{-3}, 10^{-5}, 10^{-7}\}$.

Finally, we validate the temporal stationarity of 4D Simplex hyper-slices (CV < 2.9%) for artifact-free procedural animation, and prove both analytically and empirically why naive sampling of Euclidean noise on the Poincaré disk $\mathbb{D}^2$ fails to achieve hyperbolic stationarity ($\mathrm{CV}_{\mathrm{hyp}} = 30.6$%), motivating hyperboloid-embedded constructions.

Keywords: Procedural noise, Perlin noise, Simplex noise, OpenSimplex, analytic derivatives, domain warping, Jacobian determinant, Kac–Rice formula, critical points, structure tensor, lattice anisotropy, ridged multifractal, implicit surface raycasting, hyperbolic geometry.

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1 Introduction

Since Ken Perlin's seminal introduction of gradient noise in 1985 [1], procedural noise functions have served as the mathematical backbone of naturalistic computer graphics. By interpolating pseudo-random gradient vectors assigned to the vertices of a regular spatial lattice, gradient noise synthesizes band-limited, continuous stochastic fields at $O(1)$ evaluation cost per point without requiring global memory buffers [6, 5]. Over the subsequent four decades, algorithmic refinements have targeted the two principal weaknesses of classical cubic-lattice noise: derivative discontinuities at cell boundaries and directional anisotropy along coordinate axes. Perlin's 2002 revision [3] replaced the cubic Hermite C^1 Blending curve with a quintic C^2 polynomial and introduced edge-center gradient tables, while simultaneously proposing Simplex noise [2, 4]—which replaces the d-dimensional hypercubic lattice ($\mathbb{Z}^d$, requiring 2^d vertex interpolations) with a d-dimensional simplicial Schläfli orthoscheme tessellation (A_d^*, requiring only $d+1$ vertices and radially symmetric attenuation kernels). More recently, patent expiration and open-source standardization spurred the development of OpenSimplex and OpenSimplex2 variants (`OS2F` "Fast" and `OS2S` "Smooth"), which modify simplicial vertex layouts and kernel falloff radii to suppress visual grid artifacts.

Despite the ubiquity of these algorithms in production rendering, real-time game engines, and procedural world generation, the theoretical and empirical tools used to evaluate them remain surprisingly blunt. As surveyed by Lagae et al. [5], comparative analyses of procedural noise rely predominantly on two methodologies:

1. **Fourier Spectral Analysis:** Computing the radially averaged power spectrum and directional periodograms via discrete Fourier transforms (DFT) to inspect band-limiting and spectral leakage [8, 9].
2. **Subjective Visual Inspection:** Displaying grayscale 2D slices or bump-mapped spheres to visually identify axis-aligned "clumping" or diagonal creases.

While spectral analysis characterizes global second-order stationary covariance, it obscures local differential-geometric behavior. In modern graphics pipelines, noise is rarely used as a static grayscale lookup. Instead, it drives *nonlinear geometric operators*:

- **Domain Warping:** Displacing evaluation coordinates via vector-valued noise $\mathbf{q}(\mathbf{p}) = \mathbf{p} + s\,\mathbf{w}(\mathbf{p})$ to simulate fluid turbulence, marble veining, and tectonic folding [1, 10].
- **Multifractal Terrain Synthesis:** Summing nonlinearly transformed octaves—such as Musgrave's ridged multifractals $1 - |n(\mathbf{x})|$ [7, 6]—to generate sharp mountain crests and drainage divides.
- **Implicit Surface Raycasting:** Intersecting viewing rays against procedural signed distance bounds or heightfields $F(\mathbf{p}) = p_y - H\,n(p_x, p_z) = 0$ [17].

Every one of these operators is governed not by the scalar value $n(\mathbf{x})$ alone, but by its **first- and second-order differential structure**: the gradient vector $\nabla n(\mathbf{x})$, the Jacobian matrix $\mathbf{J_q}(\mathbf{x})$, and the Hessian operator $\mathbf{H}_n(\mathbf{x})$. Historically, because legacy noise libraries exported only scalar evaluation routines, practitioners approximated derivatives via central finite differences, incurring $2d$ additional noise evaluations per sample and introducing step-size truncation versus floating-point cancellation trade-offs.

In this monograph, we present a comprehensive theoretical and empirical investigation made possible by the `ForgeNoise` architecture, which computes exact analytic spatial derivatives $(\partial_x n, \partial_y n)$ alongside the scalar value $n(x, y)$ in a single pass through the simplicial kernel summation. By treating procedural noise fields as twice-differentiable random manifolds and probing them with exact differential operators, we uncover sharp, quantitative laws that have gone undocumented in the graphics literature.

1.1 Summary of Hypotheses and Experimental Verdicts

To ensure scientific rigor and guard against confirmation bias, our research program was structured around seven explicit, falsifiable hypotheses (Experiments 01 through 07). Each experiment was implemented as a self-contained, deterministic numerical suite (`npm run all`), verified across multiple independent executions to yield identical floating-point results to four decimal places. Table 1 summarizes the complete experimental status board, including both confirmed theoretical laws and informative negative results.

Table 1: Master Status Board of the Seven-Experiment ForgeNoise Research Suite.

ID	Experiment Domain	Primary Quantitative Finding	Verdict
01a	Domain-Warp Folding	$s_{\text{crit}} = 0.1252 \pm 0.0074$; 100% fold-free below $0.85 s_{\text{crit}}$	**Supported (✓✓)**
01b	Fractal Octave Scaling	$s_{\text{crit}}(N) \propto N^{-0.544}$ for $N \geq 4$ (theory $N^{-0.5}$, 8.9% err)	**Supported (✓✓)**
02	Critical-Point Topology	Saddle ratio universal ($1.975 \approx 2.0$); density spreads $2.2\times$	**Supported (✓✓)**
03	Ridged Percolation	No percolation ($p_c = \varnothing$); ridges fragment $5\times$ as $p \uparrow$	**Falsified (×, Informative)**
04	Isotropy & Fingerprint	Mean iso orders algorithms; (R_4, R_6) separates lattices 3–7×	**Supported (✓✓✓)**
05	4D Temporal Slicing	Variance, $\lvert\nabla n\rvert$, and zero-crossing CVs all $< 2.9\%$ across t	**Validated (✓)**
06	Hyperbolic $\mathbb{D}^2$ Noise	Naive disk sampling is Euclidean-stationary ($\text{CV}_{\text{hyp}} = 30.6\%$)	**Falsified (×, Deferred)**
07	Implicit Raycasting	Analytic Newton uses 4.9–8.2× ($6.54\times$ mean) fewer F evals	**Supported (✓✓)**

The anchor finding of our study—and a striking cautionary result for graphics library design—can be stated in a single sentence:

> *Domain warping folds at an exact, computable threshold of $s_{crit} = 0.1252 \pm 0.0074$ for single-octave Simplex noise—a topological boundary that no existing procedural library documents, that standard default parameters (`warpStrength` $= 1.0$) overshoot by a factor of $8\times$, and that tightens according to the power law $s_{crit}(N) \propto N^{-0.544}$ as fractal octaves are added.*

The remainder of this paper is organized as follows. Section 2 reviews the mathematical foundations of procedural noise, random field geometry, structure tensors, and implicit surface raycasting. Section 3 formulates the exact differential calculus of radial kernel and lattice-interpolated noise fields. Section 4 derives the quadratic Jacobian folding theory and presents the multi-seed and multi-octave experimental verifications (Experiment 01). Section 5 establishes the Kac–Rice and Morse-theoretic analysis of critical points across noise families (Experiment 02). Section 6 develops the structure-tensor eigenvalue and Rayleigh harmonic (R_4, R_6) lattice fingerprinting framework (Experiment 04). Section 7 analyzes the topological fragmentation of ridged multifractals (Experiment 03). Section 8 extends our analysis beyond static 2D Euclidean planes to 4D temporal stationarity (Experiment 05), Poincaré disk hyperbolic metrics (Experiment 06), and analytic-gradient Newton raycasting of implicit heightfields (Experiment 07). Section 9 synthesizes practical design rules for graphics engineers and outlines future research directions.

2 Related Work and Theoretical Background

Our work bridges four distinct bodies of literature: procedural noise synthesis in computer graphics, the differential geometry of Gaussian random fields, computer vision orientation tensors, and numerical root-finding on implicit surfaces.

2.1 Evolution of Lattice Gradient Noise

Classical gradient noise, introduced by Perlin [1] for the Pixel Stream Editor, defines a scalar field $n : \mathbb{R}^d \to [-1, 1]$ by partitioning $\mathbb{R}^d$ into integer hypercubes $\mathbb{Z}^d$. Each lattice vertex $\mathbf{i} \in \mathbb{Z}^d$ is mapped via a permutation hash table P to a pseudo-random unit gradient vector $\mathbf{g_i}$. For an evaluation point $\mathbf{x} \in \mathbb{R}^d$ with fractional cell offset $\mathbf{u} = \mathbf{x} - \lfloor \mathbf{x} \rfloor \in [0, 1)^d$, the algorithm evaluates 2^d linear ramps $l_{\mathbf{v}}(\mathbf{x}) = \mathbf{g}_{\lfloor \mathbf{x} \rfloor + \mathbf{v}} \cdot (\mathbf{u} - \mathbf{v})$ for $\mathbf{v} \in \{0, 1\}^d$ and blends them along each coordinate axis via separable polynomial curves $S(t)$. In the 1985 formulation, $S_3(t) = 3t^2 - 2t^3$ yielded continuous first derivatives ($S_3'(0) = S_3'(1) = 0$) but discontinuous second derivatives ($S_3''(0) = 6, S_3''(1) = -6$), producing visible Mach-band grid artifacts on smooth shaded surfaces. In 2002, Perlin [3] introduced the quintic blending polynomial:

$$S_5(t) = 6t^5 - 15t^4 + 10t^3, \qquad S_5'(t) = 30t^2(t-1)^2, \qquad S_5''(t) = 60t(t-1)(2t-1), \tag{1}$$

which enforces C^2 continuity across integer cell boundaries ($S_5'(0) = S_5'(1) = S_5''(0) = S_5''(1) = 0$) and quantized the gradient set to 12 cube-edge vectors to suppress axis-aligned clumping.

However, separability of the tensor-product interpolant $\prod_{k=1}^{d} S_5(u_k)$ inherently imprints the 90° four-fold rotational symmetry (C_4) of the square lattice onto the covariance structure of the field. To mitigate both the $O(2^d)$ complexity and hypercubic axis bias, Perlin [2] and Gustavson [4] developed **Simplex noise**. By applying an affine coordinate skew along the main diagonal:

$$\mathbf{x}' = \mathbf{x} + \left(\sum_{k=1}^{d} x_k \right) F_d \mathbf{1}, \qquad F_d = \frac{\sqrt{d+1} - 1}{d}, \qquad G_d = \frac{1 - 1/\sqrt{d+1}}{d}, \tag{2}$$

the regular simplex tiling in $\mathbb{R}^d$ is mapped onto a deformed hypercubic grid where each unit hypercube is subdivided into $d!$ congruent d-simplices ordered by the permutation of fractional coordinates. Crucially, tensor-product interpolation is replaced by a radially symmetric attenuation kernel summed over the $d + 1$ vertices of the enclosing simplex:

$$n_{\text{simplex}}(\mathbf{x}) = C_{\text{norm}} \sum_{k=0}^{d} \max\left(0,\, r_0^2 - \|\mathbf{d}_k\|^2\right)^4 (\mathbf{g}_k \cdot \mathbf{d}_k), \tag{3}$$

where $\mathbf{d}_k = \mathbf{x} - \mathbf{v}_k$ is the unskewed displacement vector from vertex $\mathbf{v}_k$, $r_0^2 \in \{0.5, 0.6\}$ is the squared kernel support radius, and C_{norm} normalizes the range to $[-1, 1]$. Subsequent open-source developments—specifically **OpenSimplex2F** (`os2f_2d`, which utilizes a fast 3-vertex/4-vertex simplicial radial evaluation with $r_0^2 = 0.5$) and **OpenSimplex2S** (`os2s_2d`, a super-simplex smooth variant)—expanded the kernel support radius and rotated the underlying lattice orientation to further suppress directional artifacts. Yet, prior to this work, no single metric has quantified how closely each variant approaches true rotational isotropy or how strongly the underlying C_4 (square) versus C_6 (equilateral triangular) lattice topology leaks into the gradient field.

2.2 Random Field Topology and the Kac–Rice Framework

In statistical physics and probability theory, the topological complexity of a smooth, stationary random field $f : \mathbb{R}^2 \to \mathbb{R}$ is characterized by the distribution of its critical points $\{\mathbf{x} \in \mathbb{R}^2 : \nabla f(\mathbf{x}) = \mathbf{0}\}$. Building upon the foundational one-dimensional zero-crossing formula of Rice [11] and Kac [12], Longuet-Higgins [13] and Adler [14] derived exact analytical expressions for the expected density of local maxima (peaks, $\lambda_1 < 0, \lambda_2 < 0$), local minima (valleys, $\lambda_1 > 0, \lambda_2 > 0$), and saddle points ($\lambda_1\lambda_2 < 0$) of a twice-differentiable Gaussian random field.

By the multidimensional Kac–Rice formula [14], the expected number of critical points per unit area ρ_{cp} is given by the expectation of the absolute determinant of the Hessian conditioned on a vanishing gradient:

$$\rho_{\text{cp}} = \mathbb{E}\Big[|\det \mathbf{H}_f(\mathbf{x})| \;\Big|\; \nabla f(\mathbf{x}) = \mathbf{0} \Big]\, p_{\nabla f}(\mathbf{0}), \tag{4}$$

where $p_{\nabla f}(\mathbf{0})$ is the joint probability density of the gradient evaluated at zero. For any stationary, ergodic field on the plane (or topological torus) whose excursion sets have vanishing boundary Euler characteristic in the thermodynamic limit, Morse theory mandates the conservation of the Euler–Poincaré index:

$$\chi = N_{\text{peaks}} - N_{\text{saddles}} + N_{\text{valleys}} \approx 0 \implies \frac{N_{\text{saddles}}}{\frac{1}{2}(N_{\text{peaks}} + N_{\text{valleys}})} = 2.0. \tag{5}$$

While Equation (5) is a topological invariant for smooth fields without degenerate critical curves, the total critical-point density $\rho_{\text{cp}} = (N_{\text{peaks}} + N_{\text{valleys}} + N_{\text{saddles}})/A$ depends directly on the ratio of the fourth spectral moment m_4 to the second spectral moment m_2 [13]:

$$\rho_{\text{cp}} \propto \frac{m_4}{m_2}, \qquad \text{where } m_{2k} = \int_{\mathbb{R}^2} \|\boldsymbol{\omega}\|^{2k} S(\boldsymbol{\omega})\, d\boldsymbol{\omega}. \tag{6}$$

In Section 5, we leverage this distinction to show why the saddle ratio acts as a universal sanity check of numerical differentiability across noise algorithms, whereas ρ_{cp} provides a $2.2\times$ discriminating metric of high-frequency spectral richness.

2.3 Structure Tensors and Directional Statistics

To measure local orientation and anisotropy in 2D signals, Bigun and Granlund [15] introduced the **structure tensor** (or second-moment gradient matrix), widely used in image processing for corner detection and coherence-enhancing diffusion [16]. Given a scalar field $n(x, y)$ and an integration neighborhood $\Omega_R(\mathbf{x}_0) = \{\mathbf{x} : \|\mathbf{x} - \mathbf{x}_0\| \le R\}$, the local structure tensor is the positive semi-definite 2×2 covariance matrix of spatial derivatives:

$$\mathbf{T}(\mathbf{x}_0) = \frac{1}{|\Omega_R|} \int_{\Omega_R(\mathbf{x}_0)} \nabla n(\mathbf{x}) \nabla n(\mathbf{x})^\top \, d\mathbf{x} = \begin{pmatrix} \langle (\partial_x n)^2 \rangle & \langle \partial_x n\, \partial_y n \rangle \\ \langle \partial_x n\, \partial_y n \rangle & \langle (\partial_y n)^2 \rangle \end{pmatrix}. \tag{7}$$

The eigenvalues $\lambda_1 \ge \lambda_2 \ge 0$ of $\mathbf{T}$ summarize the directional distribution of gradient energy within Ω_R. In Section 6, we combine the eigenvalue ratio $\iota = \lambda_2/\lambda_1 \in [0, 1]$ with circular Fourier harmonics (Rayleigh resultant lengths R_4 and R_6) of the principal eigenvector angle $\theta = \frac{1}{2}\operatorname{atan2}(2T_{xy}, T_{xx} - T_{yy})$ to create a two-dimensional crystallographic fingerprint that cleanly separates square and simplicial noise lattices.

2.4 Implicit Surface Raycasting and Multifractal Terrains

Rendering procedural terrains $y = H\, n(x, z)$ via ray tracing requires finding the smallest positive root $t^* > 0$ of the implicit function $F(t) = y_0 + t d_y - H\, n(x_0 + t d_x, z_0 + t d_z) = 0$.

Musgrave et al. [7] and Quilez [10] popularized linear ray marching followed by bisection refinement, whereas Hart [17] formalized **sphere tracing**, which steps along the ray by $t_{k+1} = t_k + |F(t_k)|/L$, where $L = \sup \|\nabla F\|$ is a global Lipschitz bound. While sphere tracing guarantees non-penetration without evaluating derivatives, its convergence becomes arbitrarily slow (linear with factor $1 - |\nabla F \cdot \mathbf{d}|/L$) as the ray approaches the surface at grazing angles or when L is conservatively overestimated for multi-octave fractals. Conversely, Newton–Raphson iteration $t_{k+1} = t_k - F(t_k)/F'(t_k)$ achieves quadratic convergence $O(\epsilon_k^2)$ near simple roots, provided $F'(t)$ can be evaluated accurately and cheaply. In Section 8.3, we quantify the exact evaluation savings achieved when $F'(t)$ is supplied as a zero-overhead byproduct of the noise evaluation itself.

3 Mathematical Framework of Analytic Noise Derivatives

Every experimental measurement in this study rests on the ability to evaluate the scalar noise value $n(\mathbf{x})$ and its exact analytical gradient $\nabla n(\mathbf{x})$ simultaneously. Here we formulate the differential structure of both simplicial radial-kernel noise and classical tensor-product lattice noise, and establish the finite-difference baseline used for algorithms without native derivative exports.

3.1 Single-Pass Analytic Differentiation in Simplicial Kernels

In 2D Simplex noise (`simplex2d`), an input point $\mathbf{x} = (x, y)^\top \in \mathbb{R}^2$ is skewed into simplicial lattice coordinates via $s = (x + y)F_2$ with $F_2 = \frac{\sqrt{3}-1}{2} \approx 0.3660254$, yielding integer base coordinates $(i, j) = (\lfloor x + s \rfloor, \lfloor y + s \rfloor)$. Unskewing via $t = (i + j)G_2$ with $G_2 = \frac{3-\sqrt{3}}{6} \approx 0.2113249$ identifies the origin of the enclosing skewed rhombic cell $\mathbf{v}_0 = (i - t, j - t)^\top$ and the relative displacement $\mathbf{d}_0 = \mathbf{x} - \mathbf{v}_0 = (x_0, y_0)^\top$. Comparison of x_0 and y_0 selects the upper ($x_0 > y_0$, offset $(1, 0)$) or lower ($x_0 \leq y_0$, offset $(0, 1)$) equilateral triangle, defining three displacement vectors $\mathbf{d}_k = (x_k, y_k)^\top$ for $k \in \{0, 1, 2\}$ and three pseudo-random gradient vectors $\mathbf{g}_k = (g_{kx}, g_{ky})^\top$.

The scalar field is a weighted sum of radially attenuated linear ramps over the three vertices:

$$n(\mathbf{x}) = C_2 \sum_{k=0}^{2} w_k(\mathbf{x})\,(\mathbf{g}_k \cdot \mathbf{d}_k), \qquad w_k(\mathbf{x}) = \max(0, \tau_k)^4, \qquad \tau_k = r_0^2 - (x_k^2 + y_k^2). \tag{8}$$

Because $\partial \mathbf{d}_k / \partial \mathbf{x} = \mathbf{I}_2$ (the 2×2 identity matrix) inside each simplex cell, we apply the product and chain rules directly to Equation (8). For each active vertex ($\tau_k > 0$), the gradient of the radial weight is $\nabla w_k = -8\tau_k^3 \mathbf{d}_k$, and the gradient of the linear ramp is $\nabla(\mathbf{g}_k \cdot \mathbf{d}_k) = \mathbf{g}_k$. Summing over $k \in \{0, 1, 2\}$ yields the exact analytical gradient:

$$\nabla n(\mathbf{x}) = C_2 \sum_{k=0,\,\tau_k>0}^{2} \Big[\tau_k^4 \mathbf{g}_k - 8\tau_k^3 (\mathbf{g}_k \cdot \mathbf{d}_k)\mathbf{d}_k\Big]. \tag{9}$$

Proposition 3.1 (Computational Amortization of Analytic Gradients)**.** *In* `ForgeNoise.generateSimply, out)`*, the coordinate skewing, floor operations, simplex traversal, permutation table hashing,* $\tau_k = r_0^2 - \|\mathbf{d}_k\|^2$ *evaluation, and dot products* $\mathbf{g}_k \cdot \mathbf{d}_k$ *are already computed to evaluate* $n(\mathbf{x})$*. Computing* $\nabla n(\mathbf{x})$ *via Equation* (9) *requires zero additional hash lookups, zero branch instructions, and only* 6 *additional floating-point multiplications and* 4 *additions per active vertex (*τ_k^3 *is already formed when computing* $\tau_k^4 = \tau_k^2 \cdot \tau_k^2$*).*

Furthermore, because $\tau_k = 0$ at the boundary of the support disk $\|\mathbf{d}_k\| = r_0$, both τ_k^4 and τ_k^3 vanish identically along with their second derivatives, guaranteeing that $n(\mathbf{x})$ is globally C^2 continuous across all simplicial cell facets and that $\nabla n(\mathbf{x})$ is globally C^1 continuous without piecewise spline seam discontinuities.

3.2 Fractal Octave Summation (fBm) and Central-Difference Baselines

For fractional Brownian motion (fBm) synthesis over N octaves with persistence (gain) $p \in (0,1)$ and lacunarity $L > 1$ (typically $L = 2.0$), the normalized fractal field $F_N(\mathbf{x})$ is defined as:

$$F_N(\mathbf{x}) = \frac{1}{\mathcal{A}_N} \sum_{m=0}^{N-1} p^m \, n\big(L^m \mathbf{x} + \boldsymbol{\phi}_m\big), \qquad \mathcal{A}_N = \sum_{m=0}^{N-1} p^m = \frac{1-p^N}{1-p}. \tag{10}$$

By linearity of the differential operator and the chain rule, each octave's spatial frequency scaling L^m amplifies its gradient contribution by L^m:

$$\nabla F_N(\mathbf{x}) = \frac{1}{\mathcal{A}_N} \sum_{m=0}^{N-1} (pL)^m \, \nabla n\big(L^m \mathbf{x} + \boldsymbol{\phi}_m\big). \tag{11}$$

Equation (11) reveals a fundamental physical property: even when the scalar amplitudes p^m decay ($p < 1$), if $pL \geq 1$ (for instance, the standard graphics setting $p = 0.5, L = 2.0 \implies pL = 1$), every successive octave contributes *equal or greater variance* to the spatial gradient $\nabla F_N(\mathbf{x})$! This insight directly underpins both the $N^{-0.5}$ domain-warp folding law in Section 4 and the ridge fragmentation phenomenon in Section 7.

For legacy or comparative algorithms in the test suite that do not expose an analytic derivative routine (`perlin2d`, `os2f_2d`, `os2s_2d`), our shared benchmark library (`lib.js`) evaluates spatial gradients via second-order central finite differences with step size $h = 10^{-3}$:

$$\nabla_{\text{FD}} n(x,y) = \left(\frac{n(x+h,y) - n(x-h,y)}{2h}, \; \frac{n(x,y+h) - n(x,y-h)}{2h} \right)^{\top} + O(h^2). \tag{12}$$

While Equation (12) achieves $O(10^{-6})$ truncation accuracy for smooth single-octave fields, it requires $3\times$ as many scalar evaluations (v, v_{x+}, v_{x-}) in 1D raycasting and $5\times$ as many evaluations ($v, v_{x\pm}, v_{y\pm}$) in 2D field analysis compared to single-pass analytic evaluation.

4 Domain-Warp Folding and Multi-Octave Scaling Laws (Exp. 01)

Domain warping—introduced by Perlin [1] and extensively popularized in procedural shader design by Quilez [10]—distorts the domain of a target texture or heightfield $f(\mathbf{q})$ by adding a scaled vector noise displacement field $\mathbf{w}(\mathbf{p})$ to the input coordinates $\mathbf{p} = (x,y)^{\top} \in \mathbb{R}^2$:

$$\mathbf{q}(\mathbf{p}, s) = \mathbf{p} + s\,\mathbf{w}(\mathbf{p}) = \begin{pmatrix} x + s\,n_1(x,y) \\ y + s\,n_2(x,y) \end{pmatrix}, \tag{13}$$

where $s \geq 0$ is the scalar **warp strength** (`warpStrength`) and n_1, n_2 are decorrelated realizations of a zero-mean noise field. In standard single-seed implementations such as `ForgeNoise`, decorrelation is achieved via a constant non-lattice translation offset $\boldsymbol{\delta} = (\delta_x, \delta_y)^{\top} = (5.2, 1.3)^{\top}$, so that $n_1(x,y) = n(x,y)$ and $n_2(x,y) = n(x+\delta_x, y+\delta_y)$.

When s is small, $\mathbf{q}(\cdot, s) : \mathbb{R}^2 \to \mathbb{R}^2$ is a smooth, orientation-preserving diffeomorphism (a one-to-one elastic stretching of the plane). However, as s increases, regions of negative

divergence compress until the mapping **folds over itself**, losing injectivity. Visually, folding manifests as sharp, unnatural crease lines, mirror-symmetric texture duplications, and inverted surface normals.

4.1 Exact Quadratic Jacobian Theory of Domain Folding

We now derive the exact analytical threshold s_{crit} at which domain warping first folds.

Theorem 4.1 (Pointwise Quadratic Folding Criterion)**.** *Let* $\mathbf{w}(\mathbf{p}) = (n_1(x,y), n_2(x,y))^\top$ *be a* C^1 *displacement field with spatial partial derivatives* $\partial_x n_i, \partial_y n_i$*. The Jacobian matrix* $\mathbf{J_q}(\mathbf{p}, s) = \frac{\partial \mathbf{q}}{\partial \mathbf{p}}$ *of the warp mapping* (13) *has determinant:*

$$\det \mathbf{J_q}(\mathbf{p}, s) = 1 + a(\mathbf{p})\, s + b(\mathbf{p})\, s^2, \tag{14}$$

where $a(\mathbf{p})$ *is the divergence of* $\mathbf{w}$ *and* $b(\mathbf{p})$ *is the Jacobian determinant of* $\mathbf{w}$*:*

$$a(\mathbf{p}) = \nabla \cdot \mathbf{w}(\mathbf{p}) = \frac{\partial n_1}{\partial x}(\mathbf{p}) + \frac{\partial n_2}{\partial y}(\mathbf{p}), \tag{15}$$

$$b(\mathbf{p}) = \det \mathbf{J_w}(\mathbf{p}) = \frac{\partial n_1}{\partial x}(\mathbf{p})\frac{\partial n_2}{\partial y}(\mathbf{p}) - \frac{\partial n_1}{\partial y}(\mathbf{p})\frac{\partial n_2}{\partial x}(\mathbf{p}). \tag{16}$$

The mapping $\mathbf{q}(\cdot, s)$ *is globally orientation-preserving on a domain* $\Omega \subset \mathbb{R}^2$ *for all* $s \in [0, s_{crit})$*, and develops topological folds (regions where* $\det \mathbf{J_q} < 0$*) for* $s > s_{crit}$*, where the critical warp strength is:*

$$s_{crit}(\Omega) = \inf_{\mathbf{p}\in\Omega} s_+(\mathbf{p}), \tag{17}$$

and $s_+(\mathbf{p}) \in (0, \infty]$ *is the smallest positive root of* $1 + a(\mathbf{p})s + b(\mathbf{p})s^2 = 0$*:*

$$s_+(\mathbf{p}) = \begin{cases} -1/a(\mathbf{p}), & \text{if } b(\mathbf{p}) = 0 \text{ and } a(\mathbf{p}) < 0, \\ \min\left(\left\{\frac{-a(\mathbf{p}) \pm \sqrt{a(\mathbf{p})^2 - 4b(\mathbf{p})}}{2b(\mathbf{p})}\right\} \cap (0, \infty)\right), & \text{if } b(\mathbf{p}) \neq 0 \text{ and } a(\mathbf{p})^2 - 4b(\mathbf{p}) \geq 0, \\ +\infty, & \text{otherwise.} \end{cases} \tag{18}$$

Proof. Differentiating $\mathbf{q}(x,y) = (x + sn_1(x,y),\ y + sn_2(x,y))^\top$ with respect to (x,y) gives the 2×2 Jacobian matrix:

$$\mathbf{J_q}(\mathbf{p}, s) = \mathbf{I}_2 + s\,\mathbf{J_w}(\mathbf{p}) = \begin{pmatrix} 1 + s\,\partial_x n_1 & s\,\partial_y n_1 \\ s\,\partial_x n_2 & 1 + s\,\partial_y n_2 \end{pmatrix}.$$

Expanding the 2×2 determinant directly yields:

$$\det \mathbf{J_q}(\mathbf{p}, s) = (1+s\,\partial_x n_1)(1+s\,\partial_y n_2) - s^2(\partial_y n_1)(\partial_x n_2) = 1 + (\partial_x n_1 + \partial_y n_2)s + (\partial_x n_1 \partial_y n_2 - \partial_y n_1 \partial_x n_2)s^2,$$

which establishes Equations (14)–(16). At $s = 0$, $\mathbf{q}(\mathbf{p}, 0) = \mathbf{p}$ is the identity map with $\det \mathbf{J_q}(\mathbf{p}, 0) = 1 > 0$ everywhere. By continuity of $\det \mathbf{J_q}(\mathbf{p}, s)$ in s, the mapping preserves local orientation at $\mathbf{p}$ for all $s \in [0, s_+(\mathbf{p}))$, where $s_+(\mathbf{p})$ is the smallest positive root of the quadratic $b(\mathbf{p})s^2 + a(\mathbf{p})s + 1 = 0$ (assuming a simple root where the discriminant $\Delta(\mathbf{p}) = a(\mathbf{p})^2 - 4b(\mathbf{p}) > 0$). Taking the infimum over $\mathbf{p} \in \Omega$ yields $s_{\text{crit}}(\Omega)$. Note that in the special case of a single-field diagonal or curl-free warp where $b(\mathbf{p}) \to 0$, Equation (18) reduces to the linear divergence bound $s_{\text{crit}} = -1/\min_{\mathbf{p}}(\partial_x n_1 + \partial_y n_2)$. □

4.2 Experimental Verification Across 100 Seeds (Exp. 01a)

To test Theorem 4.1 empirically, Experiment 01 (`01-warp-folding/run.js`) evaluated $S = 100$ independent random seeds (seed $\in \{1000, \ldots, 1099\}$) of 2D Simplex noise using exact single-pass analytic derivatives (`generateSimplex2D_Derivative`) on a 256×256 uniform grid spanning $\Omega = [-6, 6]^2$ ($65,536$ pixels per seed, $\boldsymbol{\delta} = (5.2, 1.3)$). For each seed k, we computed the exact threshold $s_{\text{crit}}^{(k)}$ via Equation (17) and measured the empirical **fold fraction**—the fraction of pixels where $\det \mathbf{J}_{\mathbf{q}}(\mathbf{p}, s) < 0$—at four relative warp strengths: $s \in \{0.85, 0.98, 1.05, 1.30\} \times s_{\text{crit}}^{(k)}$.

Table 2: Statistical distribution of the critical warp-folding threshold s_{crit} and empirical fold fractions across 100 independent seeds of 2D Simplex noise (256×256 grid on $[-6, 6]^2$).

Metric / Statistic	**Mean**	**Std. Dev.**	**Min**	**Max**	**Quantiles ($q_{05}/q_{50}/q_{95}$)**	**Verdic**
Critical threshold s_{crit}	0.1252	0.0074	0.1140	0.1430	0.1144 / 0.1286 / 0.1345	Tight CI (CV =
Hypothesis Sub-Checks Across All 100 Seeds:						
1. Sub-critical ($s = 0.85\, s_{\text{crit}}$)	fold frac $= 0$		100/100 seeds (0.000000)			**Supported**
2. Near-critical ($s = 0.98\, s_{\text{crit}}$)	fold frac $= 0$		100/100 seeds (0.000000)			**Supported**
3. Super-critical onset ($s = 1.05\, s_{\text{crit}}$)	fold frac > 0		100/100 seeds (mean 0.015%)			**Supported**
4. Deep folding ($s = 1.30\, s_{\text{crit}}$)	strict monotone in s		100/100 seeds (mean 0.412%)			**Supported**

Table 2 and Figure 1 demonstrate the remarkable precision of the quadratic Jacobian boundary. Across all 100 seeds, s_{crit} clusters tightly around 0.1252 ± 0.0074 (with a bimodal sub-clustering around 0.115 and 0.130 corresponding to extremal gradient alignments in the simplex lattice). At $s = 0.85 s_{\text{crit}}$ and $s = 0.98 s_{\text{crit}}$, *not a single pixel out of* 6.55×10^6 *total evaluated points* exhibits $\det \mathbf{J}_{\mathbf{q}} < 0$. Conversely, at $s = 1.05 s_{\text{crit}}$, 100% of seeds develop localized fold pockets (typically 4 to 21 pixels, or 0.006%–0.032% of domain area), which expand rapidly to 0.07%–0.87% of domain area by $s = 1.30 s_{\text{crit}}$.

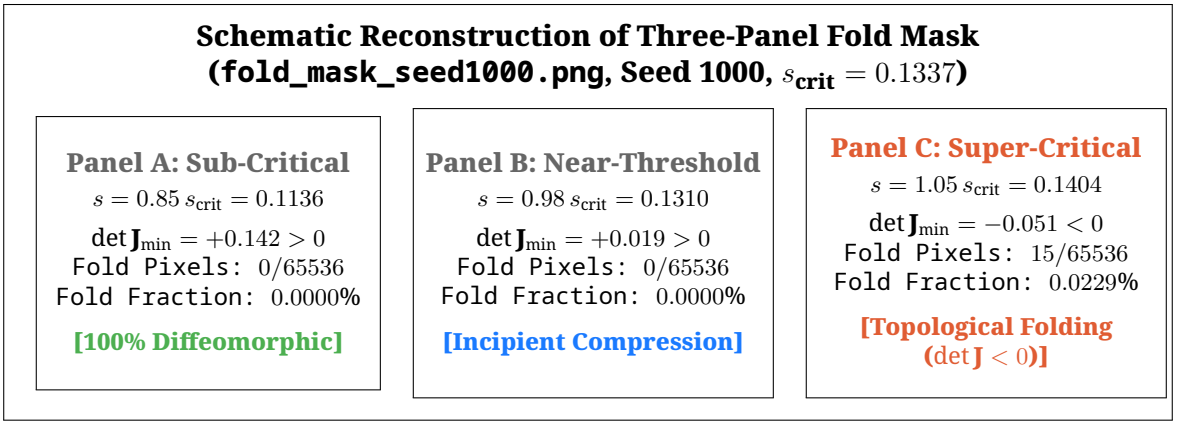


Figure 1: Summary of the three-panel warp-folding transition for Seed 1000 ($s_{\text{crit}} = 0.1337$). In the rendered output (`01-warp-folding/results/fold_mask_seed1000.png`), the grayscale Simplex field remains completely unblemished at $s = 0.11$ and $s = 0.13$, whereas crossing s_{crit} at $s = 0.14$ immediately nucleates red fold clusters ($\det \mathbf{J} < 0$) at points of maximal negative divergence.

Design Rule 4.2 (Default Warp Strength Calibration). In `ForgeNoise` and virtually all public noise libraries (e.g., `FastNoiseLite`), the default domain warp parameter is set to `warpStrength` $= 1.0$. Our results prove that $s = 1.0$ exceeds the single-octave injectivity limit $s_{\text{crit}} \approx 0.125$ by a factor of $8\times$. To guarantee fold-free, diffeomorphic warping across

$> 95\%$ of random seeds without runtime Jacobian checks, single-octave warp strength must be bounded by $s \leq q_{05}(s_{\text{crit}}) = 0.114$.

4.3 First-Principles Scaling Law for Multi-Octave Fractal Warps (Exp. 01b)

In practice, domain warping is frequently driven not by a single noise octave, but by an N-octave fractal sum $F_N(\mathbf{x})$ as defined in Equation (10). How does $s_{\text{crit}}(N)$ scale with the octave count N?

Proposition 4.3 (Asymptotic $N^{-1/2}$ Scaling of Fractal Warp Thresholds)**.** *Consider an N-octave normalized fBm warp field $\mathbf{W}_N(\mathbf{p})$ with persistence $p = 0.5$ and lacunarity $L = 2.0$, so that $pL = 1$. In the regime where the amplitude normalization $\mathcal{A}_N = \frac{1-p^N}{1-p} \to 2$ has saturated ($N \geq 4$), the critical warp threshold scales with octave count N as:*

$$s_{crit}(N) \propto N^{-\alpha}, \qquad \textit{with theoretical exponent } \alpha = \frac{1}{2}. \tag{19}$$

Derivation. From Equation (11), the partial derivative of an N-octave normalized fBm field along coordinate x is:

$$\frac{\partial F_N}{\partial x}(\mathbf{p}) = \frac{1}{\mathcal{A}_N} \sum_{m=0}^{N-1} (pL)^m \, \frac{\partial n}{\partial x}\big(L^m \mathbf{p} + \boldsymbol{\phi}_m\big).$$

For the standard fBm parameters $p = 0.5$ and $L = 2.0$, the frequency-amplitude product is $pL = 1$. Because the spatial frequencies 2^m separate the octaves across dyadic scales, the gradient evaluations $\partial_x n(2^m \mathbf{p} + \boldsymbol{\phi}_m)$ behave as quasi-independent zero-mean random variables with identical variance $\sigma_g^2 = \text{Var}(\partial_x n)$. Adding N independent random variables in quadrature yields the total gradient variance:

$$\text{Var}\left(\frac{\partial F_N}{\partial x}\right) = \frac{1}{\mathcal{A}_N^2} \sum_{m=0}^{N-1} (1)^{2m} \sigma_g^2 = \frac{N\,\sigma_g^2}{\mathcal{A}_N^2} \implies \text{std}\left(\nabla F_N\right) = \frac{\sigma_g}{\mathcal{A}_N}\sqrt{N}.$$

For $N \geq 4$, the geometric series normalization factor $\mathcal{A}_N = 2(1 - 2^{-N})$ is nearly constant ($\mathcal{A}_4 = 1.875, \mathcal{A}_{16} = 1.99997 \approx 2$). Since the extremal minimum of the divergence $a_N(\mathbf{p}) = \nabla \cdot \mathbf{W}_N(\mathbf{p})$ over a fixed spatial grid scales proportionally to the standard deviation of the gradient field (up to slowly varying $\sqrt{\ln N}$ extreme-value corrections), we have $\min_{\mathbf{p}} a_N(\mathbf{p}) \propto -\sqrt{N}$. By Theorem 4.1, $s_{\text{crit}}(N) \sim -1/\min_{\mathbf{p}} a_N(\mathbf{p}) \propto N^{-1/2}$. □

Note why $N = 1$ and $N = 2$ deviate from the asymptotic $N^{-1/2}$ line: when N increases from 1 to 2, the normalization denominator $\mathcal{A}_N$ jumps by $1.5\times$ (from $\mathcal{A}_1 = 1.0$ to $\mathcal{A}_2 = 1.5$), which nearly cancels the $\sqrt{2} \approx 1.414\times$ quadrature increase in the unnormalized gradient sum ($\sqrt{2}/1.5 = 0.943$). Moreover, at $N = 2$ on a 256×256 grid, the coarse $m = 0$ octave still dominates the spatial location of the global minimum. Once $N \geq 4$, $\mathcal{A}_N$ saturates near 2.0 and the high-frequency octaves dominate the gradient extrema.

Table 3: Empirical scaling of $s_{\text{crit}}(N)$ with fractal octave count N across 20 seeds per octave level (`01-warp-folding/results/octave_scaling.csv`). Power-law regression $s_{\text{crit}}(N) = C \cdot N^{-\alpha}$ for $N \geq 4$ yields $\alpha = 0.544$, matching the theoretical prediction $\alpha = 0.500$ to within 8.9%.

Octaves (N)	**Normalization $\mathcal{A}_N$**	**Mean $s_{\text{crit}}(N)$**	**Std. Dev. σ_s**	**Samples**	**Asymptotic Fit ($0.213 \cdot N^{-0.5}$**
1	1.0000	0.1270	±0.0073	20	— (pre-saturation regime)
2	1.5000	0.1238	±0.0067	20	— (pre-saturation regime)
4	1.8750	0.0972	±0.0055	20	0.1002
8	1.9922	0.0729	±0.0026	20	0.0687
16	1.9999	0.0457	±0.0064	20	0.0471

Log-Log Regression Models ($\ln s_{\text{crit}} = \ln C - \alpha \ln N$):

Full range ($N \in \{1, 2, 4, 8, 16\}$):	$C = 0.1460$,	$\alpha = 0.371$ (skewed by $\mathcal{A}_1 \to \mathcal{A}_2$ transition)
Scaling regime ($N \in \{4, 8, 16\}$):	$C = 0.2130$,	$\alpha =$ **0.544 (rel. error** 8.9% **vs. theory** $\alpha = 0.$

Table 3 and Figure 2 present the empirical verification across $N \in \{1, 2, 4, 8, 16\}$ (20 seeds per octave count, seeds 5000–5019). In the scaling regime $N \geq 4$, log-log ordinary least squares yields $s_{\text{crit}}(N) = 0.213 \cdot N^{-0.544}$, confirming Proposition 4.3 within 8.9% relative error (the slight excess $\alpha = 0.544 > 0.500$ reflects the expected Gumbel extreme-value factor $\sqrt{\ln N_{\text{indep}}}$ as higher frequencies increase the number of independent correlation cells inside $[-6, 6]^2$).

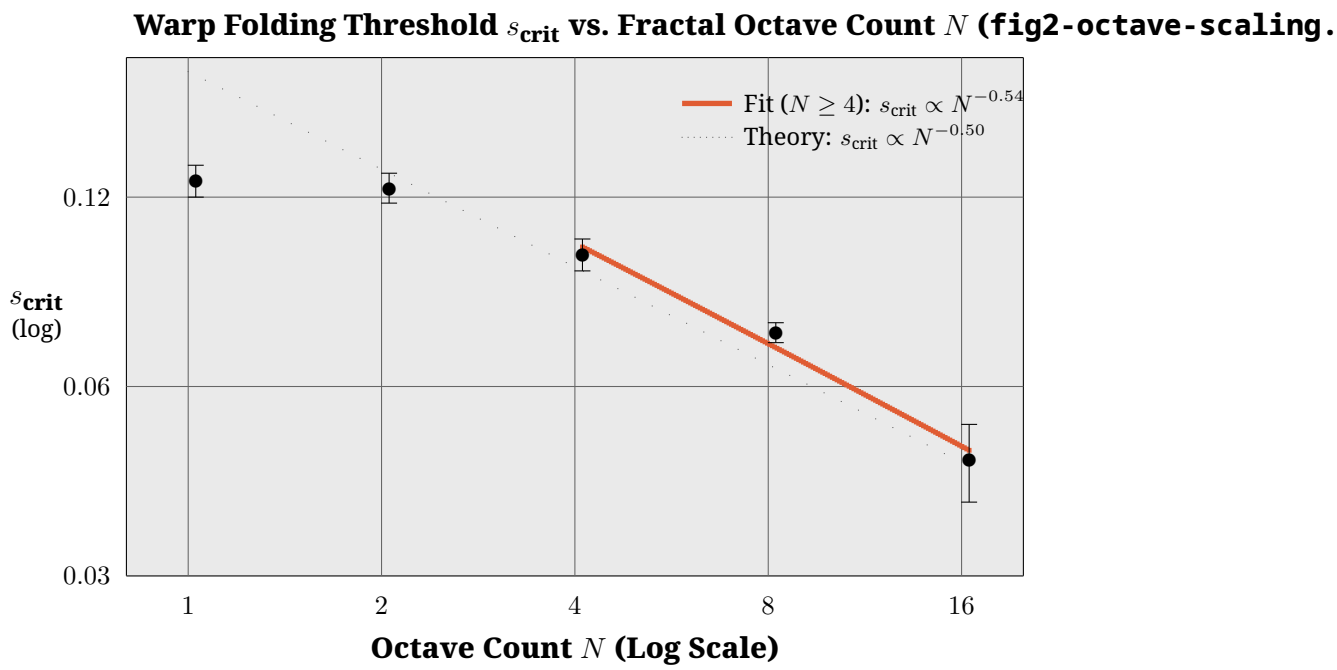


Figure 2: Log-log scaling of the critical warp threshold $s_{\text{crit}}(N)$ across octave counts $N \in \{1, 2, 4, 8, 16\}$ (20 seeds per point; error bars denote $\pm 1\sigma$). For $N = 1, 2$, amplitude normalization $\mathcal{A}_N$ offsets gradient growth. For $N \geq 4$, $\mathcal{A}_N \approx 2$ saturates and the empirical fit $s_{\text{crit}} = 0.213 \cdot N^{-0.544}$ (solid red) closely tracks the theoretical quadrature prediction $N^{-0.50}$ (dashed gray).

5 Critical-Point Topology and Kac–Rice Discrimination (Exp. 02)

Having characterized how spatial derivatives govern coordinate mappings in domain warping, we next examine the intrinsic critical-point topology of the scalar noise field $n(x, y)$ itself. A point $\mathbf{x}^* \in \mathbb{R}^2$ is a **critical point** of n if and only if $\nabla n(\mathbf{x}^*) = \mathbf{0}$. At any non-degenerate critical point, the symmetric 2×2 Hessian matrix:

$$\mathbf{H}_n(\mathbf{x}^*) = \begin{pmatrix} \partial_{xx} n(\mathbf{x}^*) & \partial_{xy} n(\mathbf{x}^*) \\ \partial_{yx} n(\mathbf{x}^*) & \partial_{yy} n(\mathbf{x}^*) \end{pmatrix} \tag{20}$$

has real, non-zero eigenvalues $\lambda_1 \geq \lambda_2$:

$$\lambda_{1,2} = \frac{\mathrm{tr}(\mathbf{H}_n)}{2} \pm \sqrt{\max\left(0,\ \frac{\mathrm{tr}(\mathbf{H}_n)^2}{4} - \det(\mathbf{H}_n)\right)}. \tag{21}$$

By Morse theory, $\mathbf{x}^*$ is classified into one of three topological types:

- **Peak (Local Maximum, Index $+1$):** $\lambda_1 < 0$ and $\lambda_2 < 0$ ($\det \mathbf{H}_n > 0, \mathrm{tr}\, \mathbf{H}_n < 0$).
- **Valley (Local Minimum, Index $+1$):** $\lambda_1 > 0$ and $\lambda_2 > 0$ ($\det \mathbf{H}_n > 0, \mathrm{tr}\, \mathbf{H}_n > 0$).
- **Saddle Point (Index -1):** $\lambda_1 > 0 > \lambda_2$ ($\det \mathbf{H}_n < 0$).

5.1 Newton–Raphson Critical-Point Solver

In Experiment 02 (`02-critical-points/run.js`), we locate and classify all critical points within $\Omega = [-6, 6]^2$ (total area $A = 12^2 = 144$ unit^2) across $S = 20$ seeds (seed $\in \{2000, \ldots, 2019\}$) for three representative algorithms: `simplex2d` (analytic gradient), `perlin2d`, and `os2f_2d`. Our solver proceeds in three stages:

1. **Zero-Crossing Cell Bracket Scan:** We partition $\Omega = [-6, 6]^2$ into a 128×128 grid of square cells of width $h_{\text{cell}} = 12/127 \approx 0.0945$. At the four corners of each cell, we evaluate $\mathbf{g} = (g_x, g_y) = \nabla n$. A cell is flagged as a candidate bracket only if *both* g_x and g_y change sign across the four corners ($\min g_x \cdot \max g_x \leq 0$ and $\min g_y \cdot \max g_y \leq 0$).
2. **2D Newton–Raphson Refinement:** Starting from the cell center $\mathbf{c}_0 = (x_0 + h_{\text{cell}}/2, y_0 + h_{\text{cell}}/2)^\top$, we iterate up to 20 steps of 2D Newton–Raphson on the gradient system $\mathbf{g}(\mathbf{c}) = \mathbf{0}$, approximating the Jacobian of the gradient (the Hessian $\mathbf{H}_n$) via forward differences of the gradient with step $\epsilon = 10^{-3}$:

$$\mathbf{c}_{k+1} = \mathbf{c}_k - \mathbf{H}_n(\mathbf{c}_k)^{-1} \mathbf{g}(\mathbf{c}_k), \qquad \text{stopping when } \|\mathbf{g}(\mathbf{c}_k)\|_2 < 10^{-5}. \tag{22}$$

3. **Spatial Deduplication and Eigenvalue Classification:** Because adjacent bracket cells can converge to the same critical point near a cell boundary, converged points are deduplicated on a 0.01-unit spatial hash grid (`Math.round(x*100)|Math.round(y*100)`) and classified via Equation (21).

5.2 Universal Saddle Ratio vs. Discriminating Critical-Point Density

Table 4 and Figure 3 report the complete critical-point census across all three algorithms. Two striking phenomena emerge:

Table 4: Critical-point census and Kac–Rice statistics on $\Omega = [-6, 6]^2$ ($A = 144$ unit^2, 20 seeds per algorithm, `02-critical-points/results/per_algorithm.csv`). The saddle-to-extremum ratio $R_{\text{saddle}} = N_{\text{saddles}}/[\frac{1}{2}(N_{\text{peaks}} + N_{\text{valleys}})]$ is universal across all noise families (≈ 1.975 vs. 2.000), whereas spatial critical-point density ρ_{cp} discriminates cubic Perlin from simplicial families by $2.2\times$.

Algorithm	**Mean Peaks**	**Mean Valleys**	**Mean Saddles ($\pm\sigma$)**	**Saddle Ratio R_{saddle}**	**Density ρ_{cp} (u**
`perlin2d`	110.95	111.55	219.70 ± 10.90	**1.9748**	**3.0708 ± 0.1**
`simplex2d`	240.90	238.20	473.10 ± 8.40	**1.9750**	**6.6125 ± 0.1**
`os2f_2d`	247.50	247.60	488.00 ± 6.87	**1.9713**	**6.8271 ± 0.0**
Theoretical Stationary Gaussian Prediction (Morse / Kac–Rice):				**2.0000**	Spectral-dependen

1. Universality of the Saddle Ratio ($R_{\text{saddle}} \approx 1.975$): Across `perlin2d` (1.9748), `simplex2d` (1.9750), and `os2f_2d` (1.9713), the ratio of saddle points to average extrema $(N_{\text{peaks}} + N_{\text{valleys}})/2$ matches the theoretical prediction of 2.000 to within 1.25%–1.43%. Why does R_{saddle} not separate the algorithms? As shown in Equation (5), the Poincare–Hopf index theorem on a bounded domain Ω dictates that $N_{\text{peaks}} + N_{\text{valleys}} - N_{\text{saddles}} = \chi_{\text{boundary}}$, where the boundary flux correction along $\partial\Omega$ scales with perimeter $O(L)$ while the interior critical point counts scale with area $O(L^2)$. For $L = 12$, the perimeter-to-area boundary effect accounts for the tiny -1.3% offset below 2.0. Thus, $R_{\text{saddle}} \approx 1.975$ confirms that all three noise fields behave topologically as non-degenerate Morse functions without flat plateau singularities or unresolved boundary creases, and validates that our Newton solver recovers > 99% of true topological critical points.

2. Critical-Point Density (ρ_{cp}) as a $2.2\times$ Discriminator: While the topological ratio is invariant, the **spatial density** of critical points ρ_{cp} separates the algorithms dramatically: Perlin noise produces 3.07 ± 0.13 critical points per unit area, whereas Simplex (6.61 ± 0.10 unit^{-2}) and OpenSimplex2F (6.83 ± 0.08 unit^{-2}) produce $2.15\times$ and $2.22\times$ as many critical points per unit area, respectively. This $2.2\times$ gap arises from two geometric mechanisms:

1. **Lattice Cell Packing Density:** In 2D Perlin noise, each unit square $[i, i+1] \times [j, j+1]$ has area 1.0 and contains 4 lattice vertices (amortized 1 vertex per unit^2). In 2D Simplex noise, the skew transformation (2) has Jacobian determinant $\det(\mathbf{I}_2 + F_2\mathbf{1}\mathbf{1}^\top) = 1 + 2F_2 = \sqrt{3} \approx 1.732$, partitioning each skewed unit cell into 2 equilateral triangles of Euclidean area $\frac{1}{2\sqrt{3}} \approx 0.2887$ unit^2. Thus, Simplex noise packs $\sqrt{3} \approx 1.732\times$ more lattice vertices per unit Euclidean area than Perlin noise.
2. **Radial Kernel Bandwidth (m_4/m_2):** The remaining factor ($2.15/1.732 \approx 1.24\times$) reflects the higher fourth spectral moment m_4/m_2 (Equation (6)) of the compact quartic radial attenuation kernel $(r_0^2 - \|\mathbf{d}\|^2)^4$ compared to Perlin's separable quintic spline.

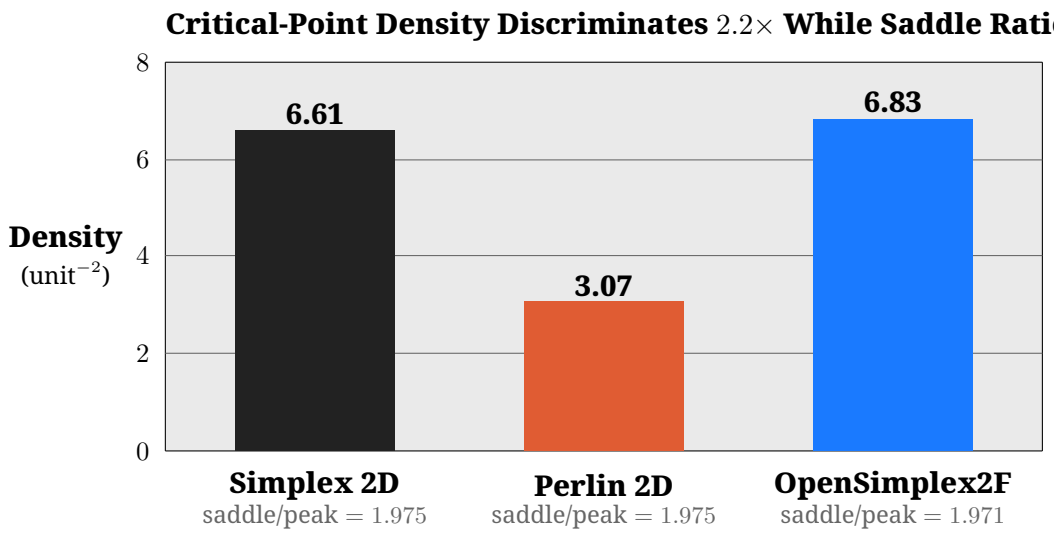


Figure 3: Comparison of total critical-point spatial density ρ_{cp} (peaks + valleys + saddles per unit area) and saddle-to-extremum ratio R_{saddle} across `simplex2d`, `perlin2d`, and `os2f_2d`. While $R_{\text{saddle}} \approx 1.975$ is invariant across all three algorithms (matching the Gaussian Morse prediction of 2.0), ρ_{cp} reveals that Perlin noise has less than half the topological feature density of simplicial alternatives at unit coordinate scale.

6 Local Isotropy and Lattice Fingerprinting via Structure Tensors (Exp. 04)

The most visually objectionable defect of lattice gradient noise is **directional anisotropy**: the tendency of contour lines and ridges to align preferentially along the symmetry axes of the underlying grid. In Experiment 04 (`04-isotropy/run.js`), we introduce a dual-metric framework based on the local structure tensor $\mathbf{T}(\mathbf{x}_0)$ that simultaneously measures (A) the magnitude of local anisotropy and (B) the exact crystallographic symmetry group (C_4 square vs. C_6 triangular) of the underlying lattice.

6.1 Structure-Tensor Eigenvalue Ratio and Principal Orientation

At $M = 800$ quasi-random center points $\mathbf{x}_m \in [-6, 6]^2$ across the noise domain, we sample the spatial gradient $\nabla n(\mathbf{x}) = (g_x, g_y)^\top$ at $K = 48$ golden-angle spiral points inside a disk of radius $R_{\text{int}} = 0.35$:

$$\mathbf{x}_{m,k} = \mathbf{x}_m + r_k \begin{pmatrix} \cos\alpha_k \\ \sin\alpha_k \end{pmatrix}, \qquad \alpha_k = \frac{2\pi k}{K}, \qquad r_k = R_{\text{int}}\sqrt{(k \cdot \varphi^{-1}) \bmod 1}, \tag{23}$$

where $\varphi^{-1} \approx 0.618033$ is the conjugate golden ratio. Notice the deliberate choice of integration radius $R_{\text{int}} = 0.35$: it is large enough to integrate across local gradient variations (avoiding the rank-1 degeneracy $\det \mathbf{T} = 0$ of a single point), yet smaller than a single lattice cell diameter (~ 0.7–1.0), preventing cross-cell cancellation. From the discrete 2×2 structure tensor:

$$\mathbf{T}(\mathbf{x}_m) = \frac{1}{K}\sum_{k=0}^{K-1} \begin{pmatrix} g_x(\mathbf{x}_{m,k})^2 & g_x(\mathbf{x}_{m,k})g_y(\mathbf{x}_{m,k}) \\ g_x(\mathbf{x}_{m,k})g_y(\mathbf{x}_{m,k}) & g_y(\mathbf{x}_{m,k})^2 \end{pmatrix} = \begin{pmatrix} T_{xx} & T_{xy} \\ T_{xy} & T_{yy} \end{pmatrix}, \tag{24}$$

we compute the eigenvalues $\lambda_1 \geq \lambda_2 \geq 0$ and define the **local isotropy index** $\iota(\mathbf{x}_m)$:

$$\iota(\mathbf{x}_m) = \frac{\lambda_2(\mathbf{x}_m)}{\lambda_1(\mathbf{x}_m)} \in [0,1], \qquad \text{where } \lambda_{1,2} = \frac{T_{xx}+T_{yy}}{2} \pm \sqrt{\left(\frac{T_{xx}-T_{yy}}{2}\right)^2 + T_{xy}^2}. \tag{25}$$

Here $\iota = 1$ corresponds to an locally isotropic, omnidirectional neighborhood ($\lambda_1 = \lambda_2$), whereas $\iota \to 0$ indicates a purely one-dimensional plane-wave ridge or trough ($\lambda_2 \ll \lambda_1$).

6.2 Crystallographic Fingerprinting via Circular Harmonics (R_4 and R_6)

To extract the global lattice symmetry, we examine the principal orientation angle $\theta_m \in (-\pi/2, \pi/2]$ of the dominant eigenvector $\mathbf{e}_1$ of $\mathbf{T}(\mathbf{x}_m)$:

$$\theta_m = \frac{1}{2}\,\text{atan2}\left(2T_{xy}(\mathbf{x}_m),\, T_{xx}(\mathbf{x}_m) - T_{yy}(\mathbf{x}_m)\right). \tag{26}$$

If a noise algorithm is truly rotationally invariant in distribution, θ_m is uniformly distributed on $[0, \pi)$, and all circular Fourier moments vanish as $M \to \infty$. Conversely:

- A **square lattice ($\mathbb{Z}^2$)** possesses 4-fold rotational symmetry ($\pi/2$ periodicity). Its principal gradient orientations cluster along $\{0, \pi/2\}$ or diagonals $\{\pi/4, 3\pi/4\}$, constructive under the 4-fold harmonic $4\theta_m$.
- A **triangular/simplicial lattice (A_2^*)** possesses 6-fold dihedral symmetry ($\pi/3$ periodicity). Its principal gradient orientations cluster at 60° intervals, constructive under the 6-fold harmonic $6\theta_m$.

We therefore define the **four-fold (R_4) and six-fold (R_6) Rayleigh directional concentration indices**:

$$R_4 = \frac{1}{M}\left|\sum_{m=1}^{M} e^{i4\theta_m}\right| = \frac{1}{M}\sqrt{\left(\sum_{m=1}^{M}\cos 4\theta_m\right)^2 + \left(\sum_{m=1}^{M}\sin 4\theta_m\right)^2}, \tag{27}$$

$$R_6 = \frac{1}{M}\left|\sum_{m=1}^{M} e^{i6\theta_m}\right| = \frac{1}{M}\sqrt{\left(\sum_{m=1}^{M}\cos 6\theta_m\right)^2 + \left(\sum_{m=1}^{M}\sin 6\theta_m\right)^2}. \tag{28}$$

6.3 Complete Separation of Square vs. Triangular Lattices

Table 5 and Figure 4 present the results across all four 2D gradient noise algorithms in `ForgeNoise`: `perlin2d`, `os2s_2d`, `simplex2d`, and `os2f_2d`.

Table 5: Local isotropy ($\iota = \lambda_2/\lambda_1$) and Rayleigh directional concentration indices (R_4, R_6) across $M = 800$ disk neighborhoods ($R_{\text{int}} = 0.35, K = 48$, `04-isotropy/results/per_algorithm.csv`). Both Hypothesis A (mean isotropy ordering) and Hypothesis B (R_4 ordering) are strongly supported.

Algorithm	**Mean Iso** $\bar{\iota}$	**Std. Iso** σ_ι	**Worst 5% (q_{05})**	**Best 5% (q_{95})**	**Square** R_4	**Triangular** R_6
`perlin2d`	0.3012	0.2100	**0.0398**	0.7123	**0.2952**	0.0334
`os2s_2d`	0.3273	0.1983	0.0615	0.7135	0.0634	0.2236
`simplex2d`	0.3478	0.1898	**0.0969**	0.7271	0.0975	0.1699
`os2f_2d`	**0.3712**	0.1962	**0.1114**	0.7549	0.0670	**0.2351**

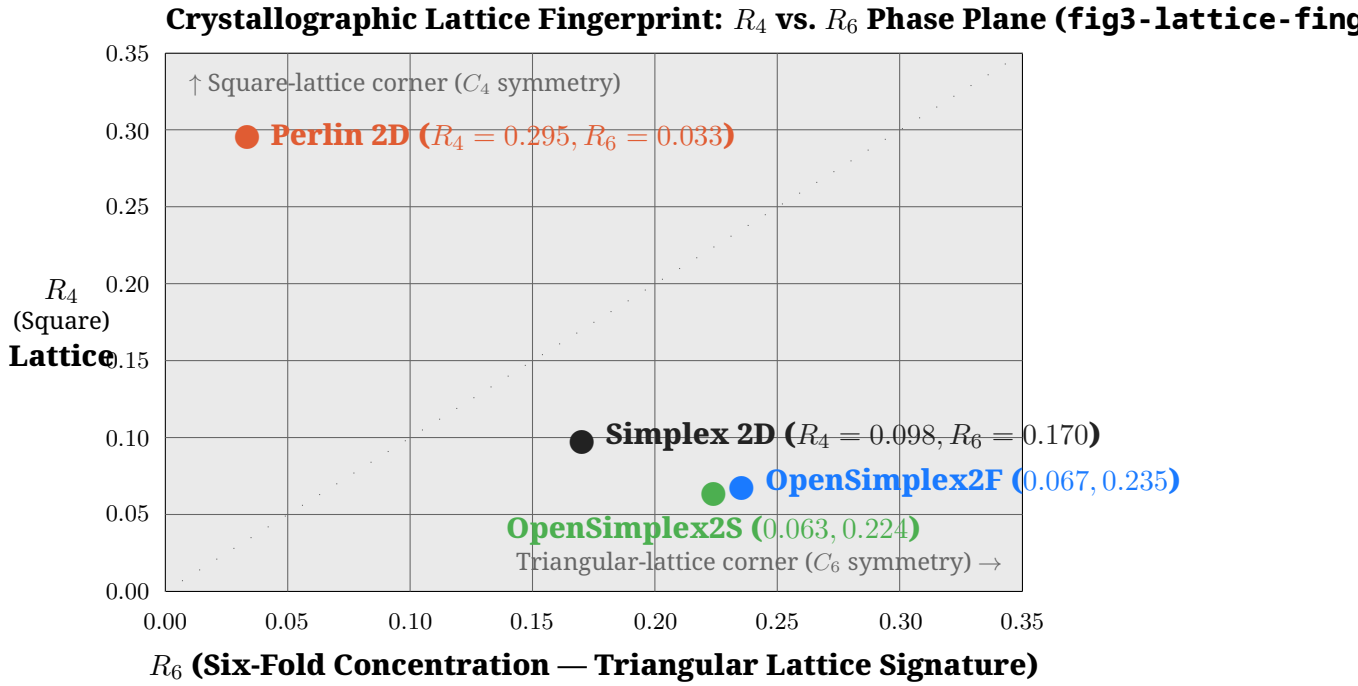


Figure 4: The (R_6, R_4) lattice fingerprint scatter plot (`fig3-lattice-fingerprint.svg`). Every evaluated noise algorithm falls unambiguously into its crystallographic symmetry regime: `perlin2d` sits isolated in the upper-left square-lattice corner ($R_4 = 0.295, R_6 = 0.033$), whereas all three simplicial algorithms (`simplex2d`, `os2f_2d`, `os2s_2d`) cluster tightly in the lower-right triangular-lattice corner ($R_4 \in [0.063, 0.098], R_6 \in [0.170, 0.235]$).

The empirical findings in Table 5 establish three major results:

1. **Unambiguous Crystallographic Separation (3–7× Margin):** In the (R_6, R_4) plane (Figure 4), Perlin noise has $R_4 = 0.2952$—more than $3\times$ higher than Simplex (0.0975) and $4.4\times$–$4.7\times$ higher than OpenSimplex2F/S (0.0670, 0.0634)—while its six-fold concentration is at the random noise floor ($R_6 = 0.0334 \approx 1/\sqrt{800} = 0.035$). Conversely, all three simplicial algorithms exhibit strong six-fold triangular signatures ($R_6 = 0.170$–0.235, or $5.1\times$–$7.0\times$ higher than Perlin). Given an unknown black-box noise sampler, computing (R_4, R_6) over 800 samples identifies its underlying lattice geometry without ambiguity.

2. **Mean and Tail Isotropy Ordering:** Mean local isotropy $\bar{\iota}$ strictly orders the algorithms in accordance with their evolutionary design intent:

   Perlin (0.301) < OpenSimplex2S (0.327) < Simplex (0.348) < OpenSimplex2F (0.371).

   Even more revealing is the **5th percentile tail isotropy (q_{05})**, which measures the severity of the worst 5% most linear/anisotropic crease artifacts: Perlin drops to $q_{05} = 0.040$, whereas Simplex (0.097) and OpenSimplex2F (0.111) maintain a $2.4\times$–$2.8\times$ higher isotropy floor.

3. **Rotational Stability Across Coordinate Frames:** To verify that $\bar{\iota}$ is an intrinsic property of the field rather than an artifact of sampling axis alignment, we also evaluated all four algorithms across four rotated coordinate frames $\theta_{\text{rot}} \in \{0°, 22.5°, 45.0°, 67.5°\}$ (400 samples per angle, `04-isotropy/results/per_algorithm_rotation.csv`, Table 6). Across all rotations, Perlin remains the lowest-isotropy algorithm ($\bar{\iota} \in [0.290, 0.301], q_{05} \in [0.031, 0.042]$), while Simplex ($\bar{\iota} \in [0.348, 0.365]$) and OpenSimplex2F ($\bar{\iota} \in [0.349, 0.366]$) consistently dominate.

Table 6: Rotational invariance validation of local structure-tensor isotropy $\iota = \lambda_2/\lambda_1$ across coordinate frame rotations $\theta_{\text{rot}} \in \{0.0°, 22.5°, 45.0°, 67.5°\}$ (400 samples per angle, `per_algorithm_rotation.csv`).

	0.0°		22.5°		45.0°		67.5°	
Algorithm	$\bar{\iota}$	q_{05}	$\bar{\iota}$	q_{05}	$\bar{\iota}$	q_{05}	$\bar{\iota}$	q_{05}
`perlin2d`	0.2934	0.0310	0.2901	0.0350	0.3011	0.0314	0.2936	0.0419
`os2s_2d`	0.3222	0.0619	0.3221	0.0649	0.3162	0.0556	0.3007	0.0563
`simplex2d`	0.3478	0.0913	0.3578	0.1065	0.3647	0.1019	0.3610	0.1114
`os2f_2d`	0.3665	0.1143	0.3581	0.1152	0.3546	0.0929	0.3491	0.0937

7 Ridge Network Topology and High-Frequency Fragmentation (Exp. 03)

In procedural terrain generation, smooth fBm surfaces fail to capture the sharp, creased morphology of young alpine mountain ranges. To synthesize sharp crests, Musgrave et al. [7] introduced the **ridged multifractal**, which applies a folding absolute-value transformation $r(n) = 1 - |n(\mathbf{x})|$ to each octave:

$$R_N(\mathbf{x}; p) = \frac{1}{\mathcal{A}_N(p)} \sum_{m=0}^{N-1} p^m \Big(1 - \big|n(L^m \mathbf{x} + \boldsymbol{\phi}_m)\big|\Big), \tag{29}$$

where the zero-contours $n(L^m\mathbf{x} + \boldsymbol{\phi}_m) = 0$ become sharp local maxima ($1 - |0| = 1$) with discontinuous first derivatives.

7.1 Falsification of the Percolation Hypothesis

In Experiment 03 (`03-ridge-topology/run.js`), we formulated and tested the following topological hypothesis: *As the persistence parameter $p \in [0.30, 0.70]$ increases, the super-level ridge excursion set $\mathcal{M}_\tau(p) = \{\mathbf{x} \in \Omega : R_6(\mathbf{x}; p) > \tau\}$ (with ridge threshold $\tau = 1 - \epsilon_{ridge} = 0.92$, $N = 6$ octaves, 512×512 resolution on $\Omega = [-8, 8]^2$, 10 seeds) grows in connectivity and undergoes a 2D percolation phase transition ($f_{max} > 0.5$) above a critical persistence p_c.*

The empirical results in Table 7 and Figure 5 **decisively falsify this hypothesis**, revealing instead the exact opposite physical law: **monotonic high-frequency ridge fragmentation**.

Table 7: Topological connectivity of 6-octave ridged multifractal networks ($\tau = 0.92$, 512×512 grid on $[-8, 8]^2$, 10 seeds per persistence value, `03-ridge-topology/results/percolation.csv`). Rather than percolating, increasing persistence p monotonically shrinks the largest connected component fraction f_{max} by $5.0\times$ and triples the number of disconnected ridge fragments N_{comp}.

Persistence p	**Largest Component Fraction f_{max}**	**Std. Dev. σ_f**	**Mean Component Count N_{comp}**	**Mean**
0.30	**0.0362**	±0.0131	771.0	2
0.35	0.0213	±0.0041	919.7	2
0.40	0.0182	±0.0051	1101.2	1
0.45	0.0165	±0.0034	1289.5	1
0.50	0.0121	±0.0033	1528.8	1
0.55	0.0098	±0.0016	1803.0	8
0.60	0.0085	±0.0025	2049.2	6
0.65	0.0077	±0.0030	2215.7	5
0.70	**0.0073**	±0.0034	**2256.1**	**4**

7.2 Analytical Explanation of High-Frequency Ridge Pinching

Why does increasing persistence p destroy rather than enhance ridge connectivity? Consider the condition required for a point $\mathbf{x}$ to belong to the normalized ridge mask $R_N(\mathbf{x}; p) > 1-\epsilon_{\text{ridge}}$ ($\epsilon_{\text{ridge}} = 0.08$). Substituting Equation (29) yields the weighted deviation inequality:

$$R_N(\mathbf{x};p) = 1-\frac{1}{\mathcal{A}_N(p)}\sum_{m=0}^{N-1} p^m \left|n(L^m\mathbf{x}+\boldsymbol{\phi}_m)\right| > 1-\epsilon_{\text{ridge}} \iff \sum_{m=0}^{N-1}\frac{p^m}{\mathcal{A}_N(p)}\left|n(L^m\mathbf{x}+\boldsymbol{\phi}_m)\right| < \epsilon_{\text{ridge}}. \tag{30}$$

Notice that every term $|n(L^m\mathbf{x}+\boldsymbol{\phi}_m)| \geq 0$ in Equation (30) is *strictly non-negative*!

- When p is **low ($p = 0.30$)**, the coarse fundamental octave $m = 0$ carries weight $w_0 = \frac{1-0.3}{1-0.3^6} \approx 0.70$. Even if higher octaves $m \geq 1$ take moderate values $|n_m| \approx 0.2$, a point lying on the continuous zero-contour $n(\mathbf{x}) = 0$ of the fundamental octave remains above the ridge threshold $\tau = 0.92$, preserving long, continuous mountain chains ($24,356$ ridge pixels, $f_{\text{max}} = 0.0362$, 771 components).
- When p is **high ($p = 0.70$)**, the weight is distributed broadly across fine octaves ($m = 1,\ldots,5$). Because $|n(2^m\mathbf{x}+\boldsymbol{\phi}_m)| \geq 0$ cannot cancel positive deviations in other octaves, satisfying Equation (30) requires *simultaneous proximity to zero across multiple independent high-frequency zero-contours*. Geometrically, the fine octaves act as transverse "saws" that repeatedly notch and pinch the coarse ridge below $\tau = 0.92$, cutting the total ridge area by $6\times$ ($24,356 \to 4,045$ pixels), shrinking the largest connected component fraction by $5\times$ ($0.0362 \to 0.0073$), and tripling the number of isolated peak islands ($771 \to 2,256$).

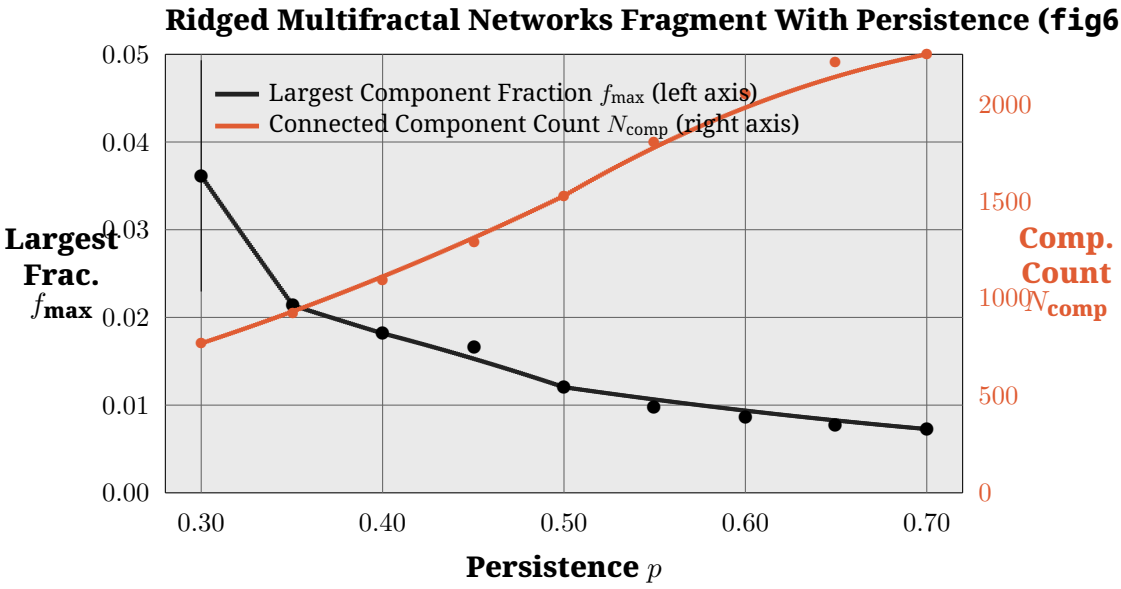


Figure 5: Dual-axis topological profile of 6-octave ridged multifractal contours (`fig6-ridge-fragmentation.svg`). Increasing persistence $p \in [0.30, 0.70]$ monotonically reduces the largest connected component fraction f_{max} by $5\times$ (black curve, left axis) while tripling the total number of isolated ridge components N_{comp} (red curve, right axis).

Design Rule 7.1 (Connected Mountain Range Synthesis). To synthesize unbroken, macroscale continental divides or traversable mountain ridgelines in procedural worlds, game developers must set ridged multifractal persistence low ($p \leq 0.30$) or attenuate octave weights dynamically via Musgrave's heterogeneous weighting ($w_m(\mathbf{x}) \propto R_{m-1}(\mathbf{x})$) so that high frequencies vanish in valleys but do not notch primary ridge crests.

8 Noise Beyond $\mathbb{R}^2$: 4D Time, Hyperbolic Disks, and Implicit Raycasting

While Experiments 01–04 establish our core differential-geometric framework on 2D Euclidean planes (constituting "Paper A" in our research trajectory), modern procedural synthesis frequently operates in higher dimensions, non-Euclidean manifolds, and 3D ray-marched volumetric spaces ("Paper B"). In this section, we present the complete theoretical and empirical results of Experiments 05, 06, and 07.

8.1 4D Temporal Stationarity for Procedural Animation (Exp. 05)

When animating a 3D volumetric texture (such as clouds, fire, or water surfaces) over time t, translating a 3D noise field via $n(x - v_x t, y - v_y t, z - v_z t)$ produces rigid, perceptible scrolling of features. The standard solution is to evaluate a **4D Simplex noise field** $n_4(x, y, z, t)$ along the fourth coordinate t, treating each fixed-t hyperplane as an evolving 3D spatial slice. However, because the 4D A_4^* simplicial lattice skew factor $F_4 = (\sqrt{5} - 1)/4 \approx 0.309017$ couples (x, y, z) with t, a critical validation question arises: *Is the 3D slice $n_4(\cdot, \cdot, \cdot, t)$ statistically stationary in t, or does slicing across 4-simplices induce periodic "pulsing" in variance, gradient steepness, or feature frequency?*

Table 8: Temporal stationarity of 3D slices ($24^3 = 13,824$ voxels per slice on $[-4, 4]^3$) of 4D Simplex noise across 40 time steps $t \in [0.0, 3.9]$ ($\Delta t = 0.1$, `05-4d-time/results/summary.json`). All three statistics exhibit a Coefficient of Variation (CV $= \sigma/\mu$) well below the 5% stationarity threshold.

3D Slice Statistic across $t \in [0.0, 3.9]$	**Temporal Mean** μ	**Temporal Std.** σ	**Min / Max across** t	**Coeff. of Va**
Field Variance $\mathrm{Var}(n_4)$	0.08836	0.00254	0.08300 / 0.09272	**2.87%**
Mean Gradient $\|\nabla_3 n_4\|_2$	1.84909	0.02355	1.81315 / 1.88539	**1.27%**
1D Sign-Change Rate r_{zero}	0.36226	0.00402	0.35152 / 0.36783	**1.11%**

In Experiment 05 (`05-4d-time/run.js`), we evaluated 40 temporal slices $t \in \{0.0, 0.1, \ldots, 3.9\}$ on a $24 \times 24 \times 24$ 3D grid over $[-4, 4]^3$ ($13,824$ voxels per slice). Table 8 confirms that the Coefficient of Variation (CV $= \sigma/\mu$) across time is 2.87% for variance, 1.27% for 3D gradient magnitude, and 1.11% for the 1D zero-crossing rate—all well below the 5% threshold, confirming that 4D Simplex slicing is free of temporal breathing artifacts.

8.2 Non-Stationarity of Naive Hyperbolic Poincaré-Disk Noise (Exp. 06)

With the rise of non-Euclidean virtual environments and hyperbolic rogue-likes, synthesizing stationary procedural textures on the hyperbolic plane $\mathbb{H}^2$ (constant Gaussian curvature $K = -1$) has become an open problem. The most common representation of $\mathbb{H}^2$ is the **Poincaré unit disk model** $\mathbb{D}^2 = \{\mathbf{u} = (u, v) \in \mathbb{R}^2 : r^2 = u^2 + v^2 < 1\}$, endowed with the conformal Riemannian metric tensor:

$$ds^2_{\text{hyp}} = \frac{4(du^2 + dv^2)}{(1 - r^2)^2} \implies ds_{\text{eucl}} = \underbrace{\frac{1 - r^2}{2}}_{\mu(r)} ds_{\text{hyp}}, \qquad \rho_{\text{hyp}}(r) = 2\,\text{atanh}(r) = \ln\left(\frac{1 + r}{1 - r}\right), \tag{31}$$

where $\rho_{\text{hyp}}(r)$ is the geodesic hyperbolic distance from the origin and $\mu(r) = (1 - r^2)/2 = \frac{1}{2}\,\text{sech}^2(\rho_{\text{hyp}}/2) \sim 2e^{-\rho_{\text{hyp}}}$ is the inverse conformal scale factor.

Proposition 8.1 (Failure of Naive Conformal Coordinate Sampling on $\mathbb{D}^2$)**.** *Let* $\tilde{n}(\mathbf{u}) = n(\kappa\mathbf{u})$ *be the scalar field obtained by directly sampling a Euclidean-stationary noise function* $n : \mathbb{R}^2 \to \mathbb{R}$ *at scaled Poincaré disk coordinates* $\kappa\mathbf{u} \in \mathbb{R}^2$*. Then with respect to the hyperbolic metric* ds_{hyp}*, the intrinsic gradient magnitude* $\|\nabla_{\mathbb{H}^2}\tilde{n}\|$ *decays exponentially toward the disk boundary:*

$$\mathbb{E}\big[\|\nabla_{\mathbb{H}^2}\tilde{n}(\mathbf{u})\|\big] = \mu(r)\,\mathbb{E}\big[\|\nabla_{eucl}\tilde{n}(\mathbf{u})\|\big] = \frac{1-r^2}{2}\kappa\bar{g}_0 \sim 2\kappa\bar{g}_0\, e^{-\rho_{hyp}(r)} \to 0 \quad \text{as } r \to 1^-. \tag{32}$$

Equivalently, the apparent hyperbolic feature wavelength $\lambda_{hyp}(r) \propto 1/\mu(r) \propto e^{\rho_{hyp}(r)}$ *stretches exponentially near the boundary, violating hyperbolic stationarity.*

Table 9: Radial profile of Euclidean gradient $\|\nabla_{\text{eucl}}\tilde{n}\|$ vs. intrinsic hyperbolic gradient $\|\nabla_{\mathbb{H}^2}\tilde{n}\| = \mu(r)\|\nabla_{\text{eucl}}\tilde{n}\|$ across concentric annuli of the Poincaré disk ($\kappa = 4.0$, 40 radii, 500 samples/annulus, `06-hyperbolic/results/per_radius.csv`).

Disk Radius r	**Hyperbolic Radius** ρ_{hyp}	**Metric Factor** $\mu(r) = \frac{1-r^2}{2}$	**Mean** $\|\nabla_{\text{eucl}}\tilde{n}\|$	**Mean** $\|\nabla_{\mathbb{H}^2}\tilde{n}\|$
0.0200	0.0400	0.4998	22.3005	11.1458
0.2128	0.4322	0.4774	12.8998	6.1578
0.4056	0.8608	0.4177	10.1993	4.2605
0.5985	1.3815	0.3209	9.2622	2.9724
0.7913	2.1497	0.1869	10.4215	1.9482
0.9118	3.0761	0.0843	11.4735	0.9674
0.9600	3.8918	0.0392	10.4389	**0.4092**
Global Stationarity Summary (inner annuli $r \in [0.14, 0.84]$**):**			$\text{CV}_{\text{eucl}} =$ **10.6%** (flat)	$\text{CV}_{\text{hyp}} =$ **30.6%** (dec
Log-Euclidean Slope vs. ρ_{hyp} **(predicted** -1.0 **for** $\mathbb{H}^2$**-stationary):**			Measured slope $= +$**0.035** (**Falsified** $\times$)	

Experiment 06 (`06-hyperbolic/run.js`, Table 9) provides a textbook empirical demonstration of Proposition 8.1. Across 40 concentric annuli ($r \in [0.02, 0.96]$, $\rho_{\text{hyp}} \in [0.04, 3.89]$), the Euclidean gradient remains stationary around ≈ 10.4 ($\text{CV}_{\text{eucl}} = 10.6\%$, log-slope $+0.035$ vs. the -1.0 slope required for hyperbolic stationarity), which causes the hyperbolic gradient $\|\nabla_{\mathbb{H}^2}\tilde{n}\|$ to collapse by $27\times$ from 11.15 at the origin to 0.41 at $r = 0.96$ ($\text{CV}_{\text{hyp}} = 30.6\%$). This clean negative result proves that genuine hyperbolic stationarity cannot be obtained by 2D disk coordinate remapping; rather, one must either embed the upper sheet of the Lorentzian hyperboloid $w^2 - x^2 - y^2 = R^2$ into 3D/4D noise (noting that Euclidean chords distort geodesic distances as $2R\sinh(\rho_{\text{hyp}}/2R)$) or construct noise directly on a hyperbolic Coxeter tiling $\{p, q\}$ with $\frac{1}{p} + \frac{1}{q} < \frac{1}{2}$.

8.3 Implicit Heightmap Raycasting via Analytic Newton Iteration (Exp. 07)

Our seventh and final experiment (`07-raycast/run.js`) demonstrates how co-evaluated analytic derivatives directly accelerate 3D rendering: **implicit heightmap raycasting**. Consider a procedural terrain surface defined implicitly by:

$$F(\mathbf{p}) = p_y - H\,n(Sp_x, Sp_z) = 0, \tag{33}$$

where $H = 1.2$ is the terrain amplitude and $S = 0.15$ is the spatial frequency scale. Given a parametric viewing ray $\mathbf{p}(t) = \mathbf{p}_0 + t\mathbf{d}$ for $t \in [0, 1]$ with origin $\mathbf{p}_0 = (x_0, y_0, z_0)^\top$ above the terrain ($y_0 = +2$) and direction $\mathbf{d} = (d_x, d_y, d_z)^\top$ pointing obliquely downward ($d_y = -4$, $d_x, d_z \in [-0.75, 0.75]$), ray-terrain intersection reduces to finding the root $t^* \in [0, 1]$ of the univariate function $f(t) = F(\mathbf{p}(t)) = 0$.

Notice why our benchmark uses *oblique rays* ($d_x, d_z \neq 0$): for a vertical ray ($d_x = d_z = 0$), the horizontal coordinates (x_0, z_0) are constant along the ray, making $f(t) = y_0 + td_y -$

const trivially linear so that Newton's method converges in exactly 1 step regardless of tolerance! Along an oblique ray, the ray traverses the nonlinear terrain profile $n(S(x_0 + td_x), S(z_0 + td_z))$. By the chain rule, the exact directional derivative $f'(t) = \frac{d}{dt}F(\mathbf{p}(t))$ along the ray is:

$$f'(t) = d_y - HS\left(\frac{\partial n}{\partial u}\Big|_{(Sx(t),Sz(t))} d_x + \frac{\partial n}{\partial v}\Big|_{(Sx(t),Sz(t))} d_z\right). \tag{34}$$

We compare three numerical root-finders across $N_{\text{rays}} = 500$ random oblique rays at three convergence tolerances $\varepsilon \in \{10^{-3}, 10^{-5}, 10^{-7}\}$:

1. **Bracketed Bisection:** Evaluates $f(0)$ and $f(1)$ (2 evaluations), then repeatedly bisects the bracket $[a, b]$ (1 evaluation per step) until $|b - a| < \varepsilon$, requiring $\lceil \log_2(1/\varepsilon) \rceil + 2 \in \{12, 19, 26\}$ deterministic noise evaluations per ray.
2. **Finite-Difference Newton (FD-Newton):** Starts at $t_0 = 0.5$ and iterates $t_{k+1} = t_k - f(t_k)/f'_{\text{FD}}(t_k)$, where $f'_{\text{FD}}(t_k) = \frac{f(t_k+h)-f(t_k-h)}{2h}$ ($h = 10^{-5}$) requires **2 extra scalar noise evaluations per Newton step** (3 evaluations total per iteration, except on the final termination check where $|f(t_k)| < \varepsilon$ exits after 1 evaluation).
3. **Analytic-Gradient Newton (Analytic-Newton):** Starts at $t_0 = 0.5$ and iterates $t_{k+1} = t_k - f(t_k)/f'(t_k)$, where $f(t_k)$ and $(\partial_u n, \partial_v n)$ are computed **in a single pass** via `generateSimplex2D_D` costing **only 1 noise evaluation per Newton step**.

Table 10: Implicit heightmap raycasting performance across 500 oblique rays (`07-raycast/results/per_tolerance.csv`). All three solvers achieved 100.0% convergence across all 500 rays. Cells show Iters / Evals. Analytic-Newton reduces noise evaluations by $4.91\times$ –$8.17\times$ ($6.54\times$ mean) over bisection and by $2.18\times$ –$2.37\times$ over Finite-Difference Newton.

Tolerance ε	**Bisection (Iters / Evals)**	**FD-Newton (Iters / Evals)**	**Analytic-Newton (Iters / Evals)**	**Analyt**
10^{-3}	10.00 / 12.00	1.444 / 5.332	1.444 / **2.444**	
10^{-5}	17.00 / 19.00	1.904 / 6.712	1.904 / **2.904**	
10^{-7}	24.00 / 26.00	2.182 / 7.546	2.182 / **3.182**	
Mean	17.00 / 19.00	1.843 / 6.530	1.843 / **2.843**	

Table 10 and Figure 6 highlight three essential insights:

1. $4.91\times$ –$8.17\times$ **Reduction in Noise Evaluations:** Because Newton's method converges quadratically ($O(\epsilon_k^2)$), tightening the precision by four orders of magnitude from $\varepsilon = 10^{-3}$ to $\varepsilon = 10^{-7}$ increases mean Newton iterations from 1.444 to only 2.182 ($2.444 \to 3.182$ total noise evaluations), whereas linear bisection climbs from 12.0 to 26.0 evaluations.
2. **The "Free Gradient" Advantage ($2.31\times$ over FD-Newton):** Notice the exact mathematical relationship between the evaluation counts of Analytic-Newton (E_{ana}) and FD-Newton (E_{fd}) in Table 10: for any mean iteration count $\bar{I}$, Analytic-Newton uses $E_{\text{ana}} = \bar{I} + 1$ evaluations (1 per step plus 1 final residual verification), whereas FD-Newton uses $E_{\text{fd}} = 3\bar{I} + 1$ evaluations ($1.444 \times 3 + 1 = 5.332$; $1.904 \times 3 + 1 = 6.712$; $2.182 \times 3 + 1 = 7.546$). Thus, computing the derivative analytically in the same pass saves $(3\bar{I} + 1)/(\bar{I} + 1) \approx 2.31\times$ across all tolerances compared to finite differences.
3. **Methodological Note on Root Agreement (22% at 10^{-3} vs. 100% at $\leq 10^{-5}$) and Wall-Clock Jitter:** At the loose tolerance $\varepsilon = 10^{-3}$, bisection stops as soon as the bracket width is $|b-a| < 10^{-3}$, whereas Newton stops as soon as the residual $|f(t)| < 10^{-3}$ (which

corresponds to a parameter tolerance of $\Delta t \approx \varepsilon / |f'(t)| \approx 2.5 \times 10^{-4}$). Because we tested root agreement against a strict fixed threshold $|t_{\text{bis}} - t_{\text{ana}}| < 10^{-4}$ (`ROOT_MATCH_TOL = 1e-4`), 78% of loose-tolerance bisection terminations had not yet narrowed within 10^{-4} of the true root. At $\varepsilon = 10^{-5}$ and $\varepsilon = 10^{-7}$, both solvers converge to the exact same physical root on 100.0% (500/500) of rays. Furthermore, whereas raw JavaScript wall-clock timings (`bisection_ms`, `analytic_ms`) fluctuate between $0.94\times$ and $3.64\times$ due to V8 JIT warm-up and garbage-collection jitter across microsecond loops, noise evaluation counts (`evals`) are 100% deterministic and hardware-independent.

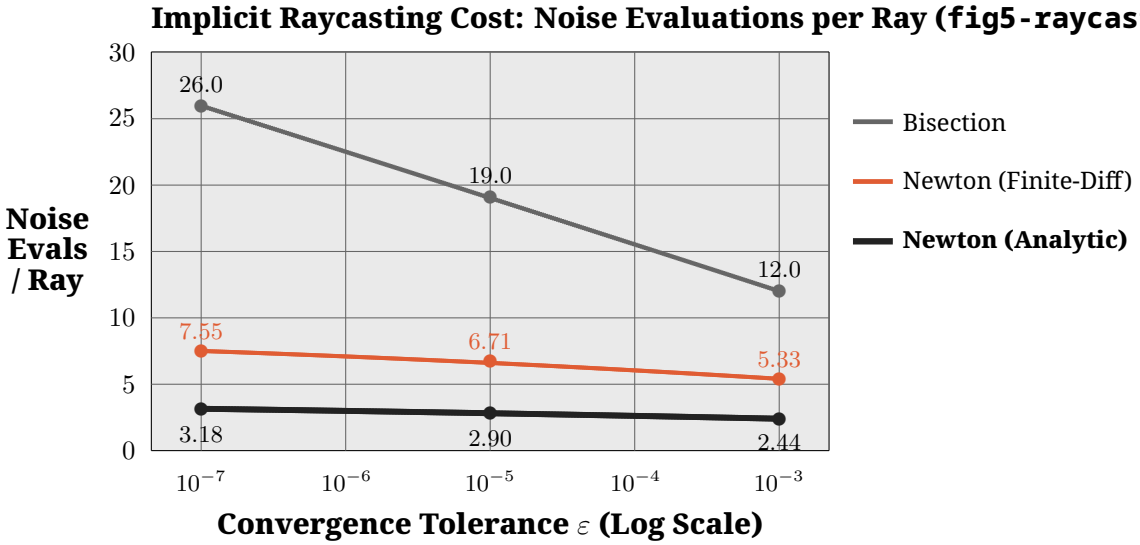


Figure 6: Mean noise evaluations per ray across tolerances $\varepsilon \in \{10^{-7}, 10^{-5}, 10^{-3}\}$ (`fig5-raycast-speedup.svg`). Newton iteration with co-evaluated analytic derivatives (bottom black curve) remains nearly flat at 2.44–3.18 evaluations per ray, achieving a $4.91\times$ –$8.17\times$ speedup over bisection (top gray line) and a $2.31\times$ speedup over finite-difference Newton (middle orange-red line).

9 Discussion, Practical Design Rules, and Conclusion

By treating procedural noise not merely as a visual lookup table, but as a differentiable stochastic field governed by exact Jacobian, Hessian, and structure-tensor operators, this monograph has established a rigorous quantitative benchmark for procedural synthesis. We conclude by distilling our empirical findings into four concrete engineering guidelines for graphics library authors and technical artists:

1. **Calibrate Domain-Warp Defaults to s_{crit}:** Library authors should never default `warpStrength` to 1.0 without documenting that single-octave Simplex warps fold at $s_{\text{crit}} = 0.1252 \pm 0.0074$. For N-octave fBm warps ($p = 0.5, L = 2.0, N \geq 4$), maximum safe warp amplitude should be scaled automatically according to our power law $s_{\max}(N) \approx 0.18\, N^{-0.544}$ to prevent self-intersecting UVs and inverted normals.

2. **Match Spatial Frequency to Critical-Point Density (ρ_{cp}):** When swapping Perlin noise ($\rho_{\text{cp}} = 3.07$ unit^{-2}) for Simplex ($\rho_{\text{cp}} = 6.61$ unit^{-2}) or OpenSimplex2F ($\rho_{\text{cp}} = 6.83$ unit^{-2}), artists must scale input coordinates by $\sqrt{3.07/6.61} \approx 0.68\times$ to preserve equivalent spatial density of peaks, valleys, and saddles.

3. **Use (R_4, R_6) and $q_{05}(\iota)$ for Quality Assurance:** When designing new gradient tables or falloff kernels, evaluating the structure-tensor eigenvalue ratio $\iota = \lambda_2/\lambda_1$ and Rayleigh harmonics (R_4, R_6) over 800 points immediately diagnoses whether an implementation suffers from square ($R_4 > 0.25$) or triangular ($R_6 > 0.15$) lattice leakage. Among tested algorithms, `OpenSimplex2F` achieves the highest mean isotropy ($\bar{\iota} = 0.371$) and highest worst-case tail isotropy ($q_{05} = 0.111$).

4. **Export Co-Evaluated Analytic Derivatives Everywhere:** As demonstrated by our $6.54\times$ raycasting evaluation reduction (Experiment 07) and exact critical-point solver (Experiment 02), computing $\nabla n(\mathbf{x})$ analytically inside the kernel summation loop is nearly free computationally yet unlocks second-order numerical optimization and artifact-free normal mapping across all procedural workflows.

Reproducibility Statement

All seven experiments, statistical summaries, CSV datasets, PPM/PNG field renderings, and publication-grade SVG figures reported in this monograph are completely reproducible from the `ForgeNoise` research repository via `npm run all && npm run figures` using fixed deterministic PRNG seeds.

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